CIE INSIDER | REVISION BOOKLET

Topic 2: Kinematics

Cambridge International AS & A Level Physics (9702) | Syllabus 2025–2027

Covers: 2.1 Equations of motion

Level: AS Level

Name: _____ Class: _____ Date: _____

How to use this booklet

1. Rate every syllabus statement Red, Amber or Green now, then again after you have worked through the booklet.
2. Draw the diagrams in Section 4 yourself. Drawing and labelling them is a skill the exam tests, so a blank box here is deliberate.
3. Answer the practice questions before you look at the mark scheme. Your teacher has the mark scheme and model diagrams.

Section 1: Syllabus checklist

Every statement in this topic is AS Level content (A Level candidates need it too).

#	Syllabus statement (verbatim)	Ref	R	A	G
2.1 Equations of motion					
1	define and use distance, displacement, speed, velocity and acceleration	2.1.1			
2	use graphical methods to represent distance, displacement, speed, velocity and acceleration	2.1.2			
3	determine displacement from the area under a velocity–time graph	2.1.3			
4	determine velocity using the gradient of a displacement–time graph	2.1.4			
5	determine acceleration using the gradient of a velocity–time graph	2.1.5			
6	derive, from the definitions of velocity and acceleration, equations that represent uniformly accelerated motion in a straight line	2.1.6			
7	solve problems using equations that represent uniformly accelerated motion in a straight line, including the motion of bodies falling in a uniform gravitational field without air resistance	2.1.7			
8	describe an experiment to determine the acceleration of free fall using a falling object	2.1.8			
9	describe and explain motion due to a uniform velocity in one direction and a uniform acceleration in a perpendicular direction	2.1.9			

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Section 2: Key definitions

Term	Definition
Distance	The total length of the path travelled (scalar)
Displacement	The distance in a straight line from a fixed point, in a stated direction (vector)
Speed	Distance travelled per unit time, or rate of change of distance (scalar)
Velocity	Change in displacement per unit time, or rate of change of displacement (vector)
Acceleration	Change in velocity per unit time, or rate of change of velocity (vector)
Uniform acceleration	Acceleration that is constant in magnitude and direction; the velocity–time graph is a straight line
Acceleration of free fall, g	The acceleration of an object falling in a uniform gravitational field with no air resistance; g = 9.81 m s^{-2} near the Earth's surface

Section 3: Formulae and equations

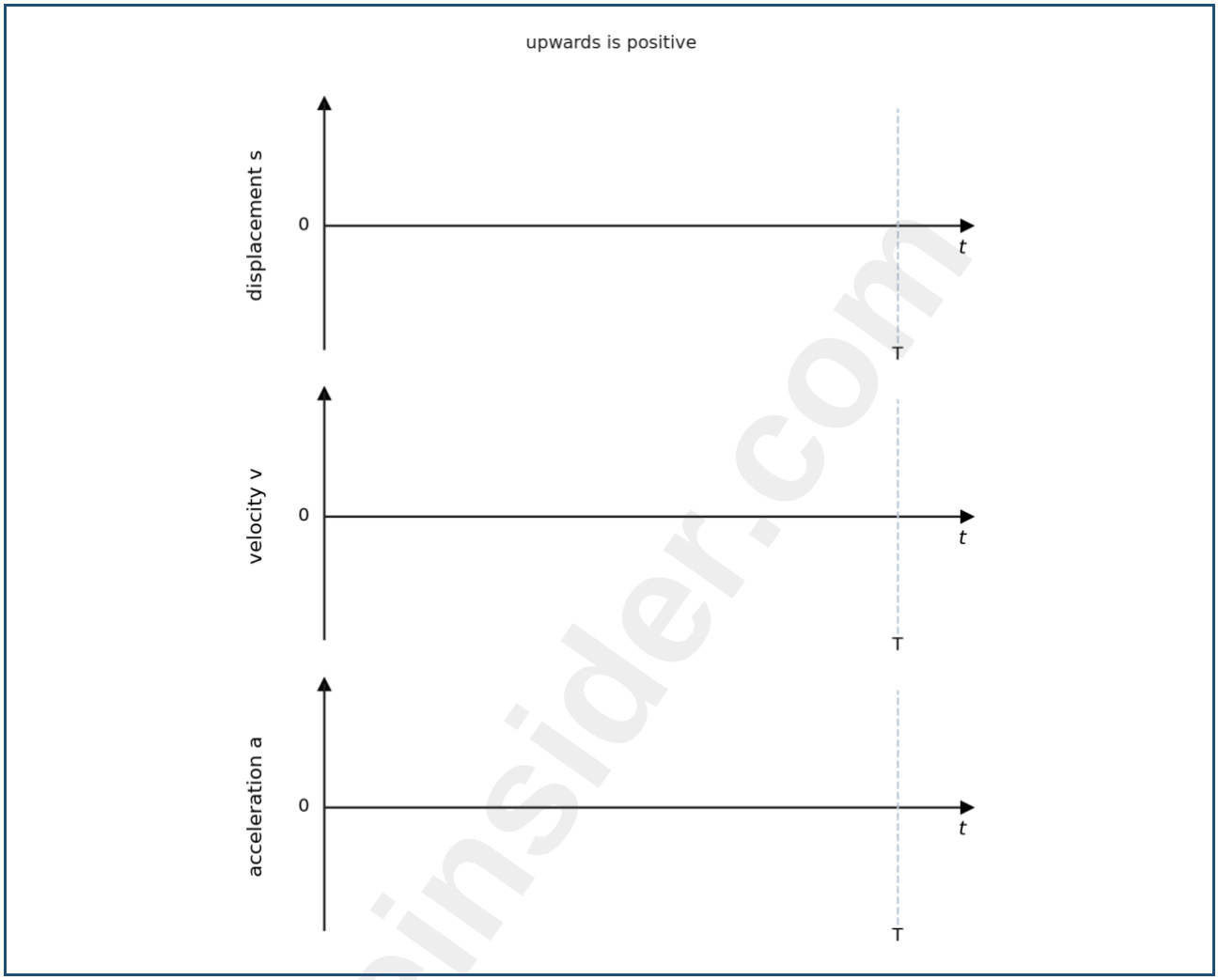
Quantity	Equation	Symbols and units
Velocity	**$v = \Delta s / \Delta t$**	Gradient of a displacement–time graph. v in m s^{-1}, s in m, t in s
Acceleration	**$a = \Delta v / \Delta t$**	Gradient of a velocity–time graph. a in m s^{-2}
Uniformly accelerated motion (recall)	**$v = u + at$**	u = initial velocity, v = final velocity, a = acceleration, t = time
Uniformly accelerated motion (recall)	**$s = ½(u + v)t$**	s = displacement. Average velocity × time
Uniformly accelerated motion (on the formulae page)	**$s = ut + ½at^2$**	Printed on the data and formulae page of Papers 1, 2 and 4
Uniformly accelerated motion (on the formulae page)	**$v^2 = u^2 + 2as$**	Printed on the data and formulae page of Papers 1, 2 and 4
Acceleration of free fall	**g = 9.81 m s^{-2}**	Given on the data page

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Section 4: Diagrams to draw

4.1 Motion graphs for a ball thrown straight up

Task: A ball is thrown vertically upwards and caught at the same height at time T. Air resistance is negligible and upwards is positive. On the three sets of axes, sketch how the displacement s, the velocity v and the acceleration a of the ball vary with time t from 0 to T. Each graph is the gradient of the one above it: use that to check your shapes.



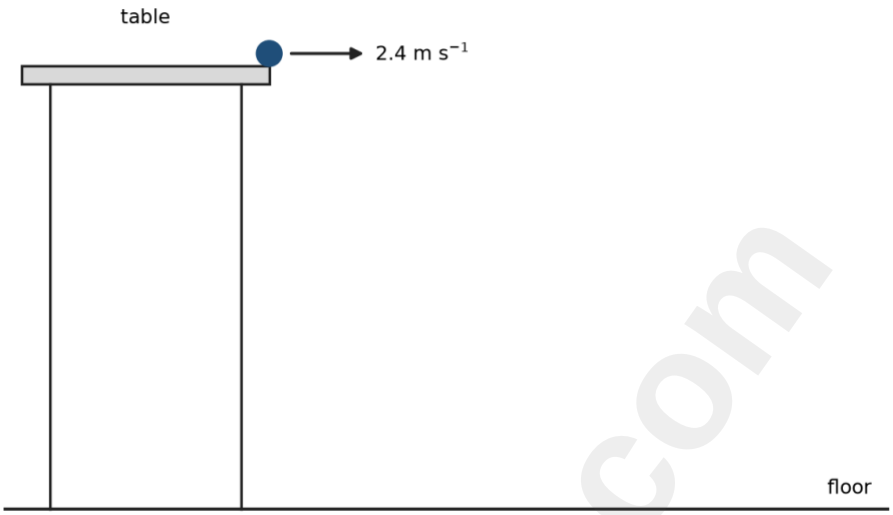
Check your diagram has:

- ☐ s: a symmetrical curve, maximum at T/2, back to zero at T
- ☐ v: a straight line starting positive, crossing zero at T/2 and ending at the same size negative
- ☐ a: a horizontal line below the time axis, the same value all the way, including at the top
- ☐ Nothing on the a graph reaches zero at the top: only v is zero there

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4.2 Path of a ball rolling off a table

Task: The ball leaves the edge of the horizontal table top at 2.4 m s^{-1}. Air resistance is negligible. Draw the path of the ball to the floor. At two points on the path, draw arrows for the horizontal and vertical components of the ball's velocity, with lengths that show how each component compares at the two points.



Check your diagram has:

- ☐ Path starts horizontally at the table edge
- ☐ Path is a smooth parabola all the way to the floor, with no vertical final section
- ☐ Horizontal component arrows the same length at both points
- ☐ Vertical component arrow longer at the later point
- ☐ Arrows labelled horizontal and vertical

4.3 The four equations of motion

Task: Complete the table. For each equation, write the one quantity out of s, u, v, a and t that it does not contain, and whether it is printed on the formulae page.

Equation	Quantity it does not contain	Printed on the formulae page?
$v = u + at$		
$s = ½(u + v)t$		
$s = ut + ½at^2$		
$v^2 = u^2 + 2as$		

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Section 5: Worked examples

Worked Example 1 AS Level

Starting from the definitions of velocity and acceleration, derive $v = u + at$ and $s = \frac{1}{2}(u + v)t$ for an object moving in a straight line with uniform acceleration a.

Acceleration is change in velocity per unit time. The velocity changes from u to v in time t, so $a = (v - u) / t$.

Rearrange: $at = v - u$, so $v = u + at$.

The acceleration is uniform, so the velocity–time graph is a straight line and the average velocity is $\frac{1}{2}(u + v)$.

Velocity is change in displacement per unit time, so displacement = average velocity × time.

$v = u + at$ and $s = \frac{1}{2}(u + v)t$

Worked Example 2 AS Level

A ball is thrown vertically upwards at 14.0 m s^{-1} from the edge of a cliff 25.0 m above the sea. Air resistance is negligible. Calculate the time the ball takes to reach the sea.

Take upwards as positive. $u = +14.0$ m s^{-1}, $a = -9.81$ m s^{-2}, and $s = -25.0$ m because the sea is below the starting point.

$s = ut + \frac{1}{2}at^2$: $-25.0 = 14.0t - 4.905t^2$

Rearrange: $4.905t^2 - 14.0t - 25.0 = 0$

$t = [14.0 + \sqrt{(14.0^2 + 4 \times 4.905 \times 25.0)}] / (2 \times 4.905) = (14.0 + 26.20) / 9.81$

The other root is negative, so it is rejected.

t = 4.10 s

Worked Example 3 AS Level

A ball is kicked from level ground at 18.0 m s^{-1} at 35° above the horizontal. Air resistance is negligible. Calculate the time of flight and the horizontal distance travelled before the ball lands.

Resolve the initial velocity. Vertical: 18.0 sin 35° = 10.32 m s^{-1}. Horizontal: 18.0 cos 35° = 14.74 m s^{-1}.

Vertically (upwards positive), the ball lands when $s = 0$: $0 = 10.32t - \frac{1}{2} \times 9.81 \times t^2$, so $t = 2 \times 10.32 / 9.81 = 2.10$ s

Horizontally there is no acceleration: distance = 14.74 × 2.105

time of flight = 2.10 s; horizontal distance = 31.0 m

Section 6: Multiple choice

Paper 1 style. Choose one answer, A, B, C or D. 8 marks, one for each question.

1. Which statement defines acceleration?

A change in velocity per unit time

B change in speed per unit time

C rate of change of velocity per unit time

D velocity per unit time

Answer: _____

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2. A cyclist rides 400 m due north and then 300 m due east. The whole journey takes 100 s. What is the magnitude of the average velocity of the cyclist?

A 1.0 m s^{-1}
B 4.0 m s^{-1}
C 5.0 m s^{-1}
D 7.0 m s^{-1}

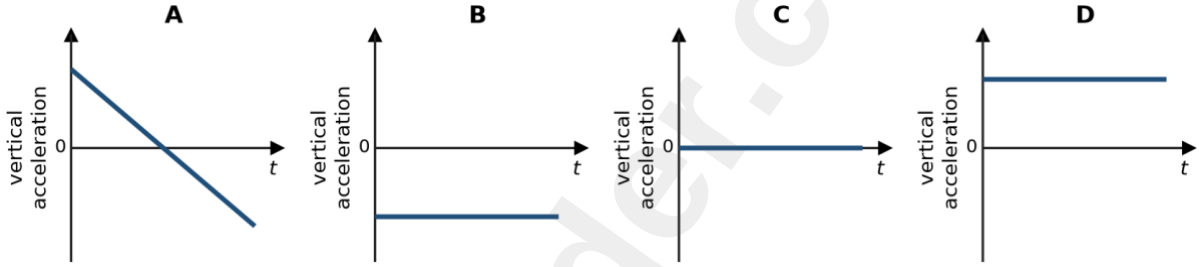
Answer: _____

3. The velocity of a car decreases uniformly from 25 m s^{-1} to 5.0 m s^{-1} in 4.0 s. What distance does the car travel in this time?

A 5.0 m
B 20 m
C 40 m
D 60 m

Answer: _____

4. A ball is thrown at an angle above the horizontal. Air resistance is negligible and upwards is positive. Which graph shows how the vertical component of the acceleration of the ball varies with time t while it is in the air?



Answer: _____

5. A stone is thrown horizontally at 12 m s^{-1} from the top of a vertical cliff 45 m high. Air resistance is negligible. How long does the stone take to reach the sea at the bottom of the cliff?

A 2.0 s
B 3.0 s
C 3.8 s
D 4.6 s

Answer: _____

6. A ball is thrown vertically upwards at 15.0 m s^{-1}. Air resistance is negligible and upwards is positive. What is the velocity of the ball 2.00 s later?

A −4.62 m s^{-1}
B 4.62 m s^{-1}
C 19.6 m s^{-1}
D 34.6 m s^{-1}

Answer: _____

7. A ball is kicked at 20 m s^{-1} at 30° above the horizontal. Air resistance is negligible. What is the speed of the ball at its highest point?

A 0
B 10 m s^{-1}
C 17 m s^{-1}
D 20 m s^{-1}

Answer: _____

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8. A student measures the time t for a steel ball to fall from rest through different heights h. Which graph gives a straight line with a gradient equal to g / 2?

A h against t
B t against h
C h against $1 / t^2$
D h against t^2

Answer: _____

Section 7: Structured questions

Paper 2 style. Total: 55 marks. Show your working in every calculation.

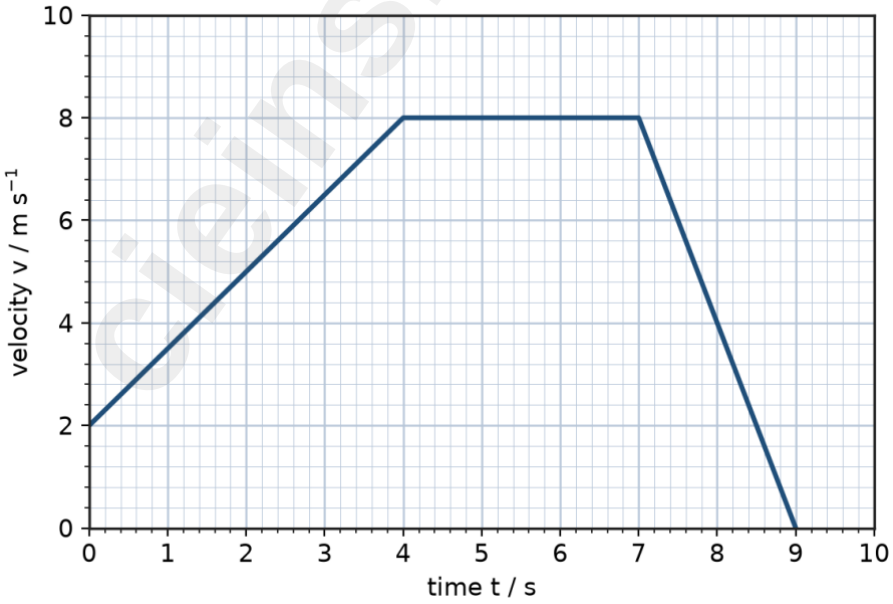
1. Kinematics definitions.

(a) Define velocity. [1]

(b) Define acceleration. [1]

(c) State one difference between distance and displacement. [1]

2. The graph shows how the velocity v of a trolley varies with time t.



(a) Calculate the acceleration of the trolley from t = 0 to t = 4.0 s. [2]

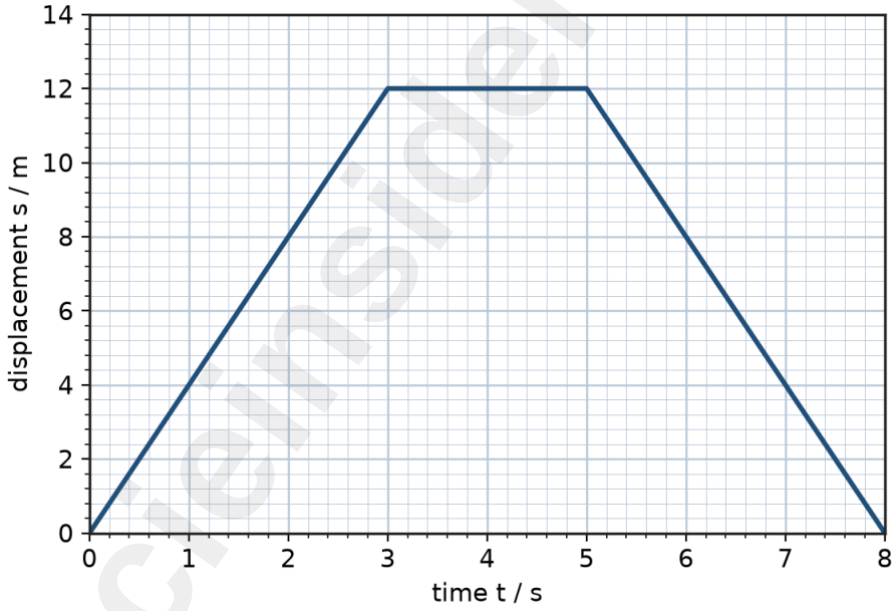
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(b) Determine the displacement of the trolley from t = 0 to t = 9.0 s. [3]

(c) Calculate the acceleration of the trolley from t = 7.0 s to t = 9.0 s. [2]

(d) Calculate the average velocity of the trolley from t = 0 to t = 9.0 s. [1]

3. The graph shows how the displacement s of a toy car from its starting point varies with time t. The car moves along a straight line.



(a) Determine the velocity of the car from t = 0 to t = 3.0 s. [1]

(b) Determine the velocity of the car from t = 5.0 s to t = 8.0 s. State what the sign of your answer shows. [2]

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(c) For the whole 8.0 s, state the distance travelled and the displacement at t = 8.0 s. [2]

(d) Calculate the average speed of the car for the whole 8.0 s. [1]

4. An object moves in a straight line with uniform acceleration a. Its velocity increases from u to v in time t.

(a) Sketch the velocity–time graph for this motion. Use the area under your graph to show that the displacement $s = ut + \frac{1}{2}at^2$. [3]

(b) Use $s = \frac{1}{2}(u + v)t$ and $v = u + at$ to show that $v^2 = u^2 + 2as$. [2]

5. A car travelling at 27 m s^{-1} brakes with uniform deceleration and stops in a distance of 54 m.

(a) Calculate the deceleration of the car. [2]

(b) Calculate the time the car takes to stop. [2]

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(c) The driver's reaction time is 0.80 s. The car travels at constant speed during this time. Calculate the total stopping distance from the moment the driver sees a hazard. [2]

6. A stone is thrown vertically upwards at 12.0 m s^{-1} from a bridge. The point of release is 20.0 m above the water. Air resistance is negligible.

(a) Calculate the maximum height of the stone above the point of release. [2]

(b) Calculate the time from release until the stone hits the water. [3]

(c) Calculate the speed of the stone as it hits the water. [2]

(d) A student says that the acceleration of the stone is zero at its highest point. Comment on this statement. [1]

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7. Describe an experiment to determine the acceleration of free fall g using a falling object. Include the measurements you would take, how you would make them accurate, and how you would find g from a graph. [5]

8. A ball is kicked from level ground at 16.0 m s^{-1} at 40° above the horizontal. Air resistance is negligible.

(a) Calculate the horizontal and vertical components of the initial velocity. [2]

(b) Calculate the time taken for the ball to reach its highest point. [1]

(c) Calculate the maximum height reached by the ball. [2]

(d) Calculate the horizontal distance from the kick to where the ball lands. [2]

(e) State the horizontal and vertical components of the acceleration of the ball at its highest point. [1]

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9. A ball rolls off the edge of a horizontal table top at 2.4 m s^{-1}. The table top is 1.25 m above the floor. Air resistance is negligible.

(a) Calculate the time the ball takes to reach the floor. [2]

(b) Calculate the horizontal distance from the table edge to where the ball lands. [1]

(c) Calculate the speed of the ball just before it hits the floor. [3]

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Section 8: Examiner tips

Where a tip comes from a Cambridge examiner report for 9702, the exam series and paper are given in brackets.

- Define velocity as 'rate of change of displacement' or 'change in displacement per unit time', never both at once. The June 2022 report lists 'the rate of change of displacement per unit time' as a wrong definition. (June 2022, Paper 21)
- Define acceleration as 'change in velocity per unit time' or 'rate of change of velocity'. Reports have rejected 'velocity / time', and using 'speed' in place of velocity. (March 2024, Paper 22; June 2025, Paper 21)
- Displacement is the area under a velocity–time graph. Split the area into rectangles and triangles. If the line does not start at zero velocity, the shape is a trapezium: in March 2021 a significant minority of candidates used a triangle instead. (March 2021, Paper 22)
- Before you use an equation of motion, check whether the object really starts from rest. Substituting $u = 0$ for an object that was already moving is a recorded error, in March 2021 and again in Nov 2022. (March 2021, Paper 22; Nov 2022, Paper 23)
- Choose a positive direction and write it down. If upwards is positive, $g = -9.81\ \text{m s}^{-2}$ for the whole flight, on the way up and on the way down. Reports record +9.81 substituted for an object moving upwards, and inconsistent signs for s and g. (June 2024, Paper 21; June 2025, Paper 21)
- In projectile motion, treat the horizontal and vertical motions separately. The horizontal velocity is constant, so there is no horizontal acceleration: never use $9.81\ \text{m s}^{-2}$ horizontally, and never divide a vertical distance by a horizontal speed. (June 2024, Paper 21; Nov 2024, Paper 23)
- Sketching the path of an object released horizontally with no air resistance: start the curve horizontally and keep it a parabola all the way down. In Nov 2024 many candidates drew a parabola with a vertical final section even though there was no air resistance. (Nov 2024, Paper 23)
- Only two equations of motion are printed on the formulae page: $s = ut + \frac{1}{2}at^2$ and $v^2 = u^2 + 2as$. Learn $v = u + at$ and $s = \frac{1}{2}(u + v)t$, and be able to derive all four from the definitions of velocity and acceleration.
- To find g with a falling object, plot h against t^2. For an object released from rest $h = \frac{1}{2}gt^2$, so the gradient is $g / 2$ and $g = 2 \times$ gradient.

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